A Claim-Admission State Machine for Diffusion Membership-Inference Scores: Boundary Cases and Evidence States

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Abstract—**Diffusion membership-inference audits need more than scalar attack scores. In our protocol, a claim is reportable only when its packet names the target, split, per-sample observable, metric provenance, finite-tail basis, and consumer use boundary. Admission is decided by a finite-state rule: each proposed audit claim is checked against its packet, and a failed claim names the first missing requirement. We exercise the rule on frozen boundary cases as protocol checks: three protocol-admitted replay rows, two source-documented point rows, one repeated-response candidate with AUC 0.9615, one MoFit public score-file boundary case with AUC 0.941948, and a separate C14 packet of thirteen internal public-surface stress rows. MoFit remains support-only because target identity, explicit row IDs, certified row binding, consumer boundary, and surface delta are incomplete. C14 remains packet-ready only; it is not external agreement, field prevalence, coding reliability, or admitted score/response evidence. The author-keyed summary still exposes the measurement failure mode: code and paper-artifact links are present in 12/13 selected rows, while no selected row passes the score/response or consumer-boundary gate. The application treats diffusion MIA as claim-bound measurement: supported leakage claims are retained, while signals that lack required surfaces stay candidate, support-only, or negative and cannot become downstream audit claims or field-rate conclusions.**

I. INTRODUCTION

Diffusion models can memorize or replicate training examples in ways that enable membership inference attacks [1], [2]. The literature has produced black-box, gray-box, and white-box signals, including reconstruction similarity, posterior-error statistics, proximal initialization, conditional likelihood discrepancy, black-box reconstruction, and white-box gradient attacks [3], [4], [5], [6], [7]. However, aggregate summary scores are not automatically audit evidence. We study a reporting risk in diffusion MIA work: claim escalation, where a score, partial artifact, or candidate signal is reported as a stronger downstream audit claim than the evidence packet supports. A model owner, replay auditor, or downstream report consumer needs to know which target checkpoint was audited, what row IDs define members and nonmembers, whether scores exist per sample, how low-FPR tails should be interpreted, and which claims are blocked by missing artifacts.

We encode this gap as a claim-bound evidence packet. The audit object is the structured packet with defined surfaces, not a method name or a single score. Following security-measurement and reproducibility norms [8], [9], [10], the contract specifies target identity, split semantics, score or response coverage, metric provenance, consumer boundary, and surface delta as gates. Given frozen gate labels, a rule maps each claim to an audit state: a reader can inspect the gate labels in the claim register, identify the first missing surface, and check whether the allowed wording would change. This differs from attack benchmarking: the measured object is claim-support eligibility, not attack-method ranking. The study then identifies the audit wording each concrete packet can support under a fixed evidence contract.

We evaluate three packet questions: which fields make a diffusion MIA result admissible for model owners, replay auditors, and downstream reports; which existing score or response packets expose those fields; and which high-score or metadata rows fail because a required surface is absent. The answers bound allowed audit claims instead of producing attack ranks. Under this contract, AUC magnitude and metadata-only records do not make a row admitted unless the consumer-boundary and surface-delta evidence required by the claim is present. Such rows remain candidate, support-only, or blocked until the missing surface is supplied.

The paper contributes:

- A claim-admission state machine with six gated surfaces: target identity, split semantics, score/response coverage, metric provenance, consumer boundary, and surface delta. The transition rule maps a proposed audit wording to reportable, candidate, support-only, watch, or negative states by recording the first missing surface and the narrower wording the packet supports.
- We apply the state machine to replay positives, point comparators, metric-only negatives, and source/artifact stops, then record the missing surfaces behind interval, dominance, and leaderboard claims.
- A repeated-response candidate (AUC 0.9615) is non-admitted when consumer-boundary and surface-delta gates are missing; same-family metric checks alone do not admit the row.
- A public score-file boundary case (MoFit, AUC 0.941948) remains support-only when public score-like files lack immutable target identity, explicit row IDs, certified row binding, consumer-boundary support, and surface delta.
- The paper also reports two preparation checks: a thirteen-row C14 stress packet and ten selected renderer and phrase-guard sanity cases. Before reviewer labels, C14 shows the weak-rule failure mode: code and paper-artifact links are present in 12/13 selected rows, while no selected row passes the score/response or consumer-boundary

gate. Both checks are packet-ready controls only and do not support adjudication, prevalence, reliability, deployment, or compute-release claims.

II. Related Work

A. Membership Inference and Evidence

Membership inference asks whether a record was used during training [11]. Prior work links this risk to overfitting and to the statistical meaning of membership scores [12], [13], while recent position work warns against treating MIA as standalone proof of an individual datum's use [14]. Prior work explains why MIA scores alone do not prove training-data use. It does not specify how auditors should scope a score-backed claim. Foundation-model audits also show that blind baselines can outperform MIA attacks when member and nonmember sources differ for reasons unrelated to membership [15]. We specify a claim-admission protocol: target, split, row-level observable, metric provenance, and consumer boundary must be explicit before wording can strengthen.

B. Security Measurement and Audit Claims

Security/privacy measurement differs from attack benchmarking because the measured unit, denominator, replay surface, and downstream consumer determine what a number can mean. Security-science and benchmarking work similarly argues that scientific security claims require explicit evidence units, denominators, measurement assumptions, and reproducible comparisons [8], [9]. For MIA, first-principles metric analysis shows that accuracy and tail rates require careful base-rate and threshold interpretation [13], while recent audit-oriented interpretations warn that membership evidence is not the same as proof of training-data use [14]. This paper specifies a claim-admission protocol for diffusion MIA: it records which target/split/row observable and replay tier make a score consumable, and it preserves partial packets as candidate or support evidence without forcing them into a leaderboard. Attack papers provide evidence surfaces and packets; our object is to assign claim-support states under a fixed contract, not to improve or rank attack methods.

C. Diffusion Memorization and MIA

Diffusion models [16], [17] expose privacy risk through memorized generations and membership signals [1], [2]. The access surface matters: gray-box posterior, trajectory, and likelihood surrogates [3], [4], [5], [18], black-box response or reconstruction methods [19], [6], caption-free fitted-embedding scores [20], and white-box gradient attacks [7] support different audit claims. Recent initial-noise probing expands this taxonomy to noise-schedule surfaces, reinforcing that modality-specific signals should be gated separately [21]. DPDM, CopyMark, MIDST, and CDI further show that defense, benchmark, and dataset-level contracts change what a result can mean [22], [23], [24], [25].

D. Reporting and Reproducibility Artifacts

Reproducibility checklists, datasheets, and model cards document data, code, conditions, origins, limitations, and intended use [10], [26], [27]. We measure sentence admissibility under a row-bound packet. Row-level membership labels or responses, metric provenance, finite-tail denominators, replay tier, surface delta, and allowed consumer language jointly decide what a number can mean. Two artifacts with similar code availability can receive different claim states if only one exposes row-level provenance, finite-tail basis, or a consumer action path for the claim. An incomplete artifact is not automatically a failed reproduction, but the audit claim it can support is bounded by what its packet exposes.

III. Audit Surfaces

This paper studies audit admission: what a score packet can support when it does not prove individual training-data membership from scores alone. The membership game is binary: for a fixed target and frozen member/nonmember split, an evidence packet scores the fixed benchmark member/nonmember label under a stated target, split, and access mode. The audit output is the claim this packet can support. Query budgets, sample counts, and auxiliary access are packet-specific and are not normalized into a universal leaderboard. We explicitly exclude individual training-data proof, field-wide artifact prevalence, and exposing candidate rows as consumer-facing audit-report claims. The split gate therefore requires membership semantics, not merely two convenient source pools.

We group prior diffusion MIA artifacts by access surface instead of headline score [3], [4], [5], [19], [6], [7]. Diffusion MIA results observe different surfaces and are therefore not directly interchangeable. A black-box response packet can support an output-consumer claim if target identity and query rows are fixed. A gray-box trajectory or posterior-statistics packet can support a mechanism or evaluator claim, but requires a different consumer boundary. A white-box gradient packet is useful as an upper-bound risk comparator, not as a black-box report. Feature packets and separately supplied local scores can support mechanism analysis or hypothesis generation while remaining support-only as audit evidence. Table I records the surfaces used in this paper. Access mode is not interchangeable: a row reported under one surface is not automatically reportable under another.

IV. Evidence Contract

The protocol uses six gates to decide whether a result can support a portability or downstream audit claim; tested portability routes fail on current packets. The target identity gate requires a fixed model architecture, checkpoint identity, training provenance, and accessibility. The split contract requires precise member and nonmember manifests. The score or response gate requires row-level coverage and labels. The metric gate requires AUC, ASR, TPR@1%FPR, TPR@0.1%FPR, nonmember denominator, and a declared replay tier: row-score replay, response/feature replay, target-score replay, or

TABLE I
AUDIT SURFACES AND ALLOWED EVIDENCE ROLES.

Surface	Allowed paper role
Black-box responses	Reportable only with fixed target, split, response coverage, and metric replay.
Gray-box trajectory/statistics	Reportable or candidate depending on provenance and consumer semantics.
White-box gradients	Upper-bound comparator unless the audit consumer has white-box access.
Feature packets	Support-only unless raw target/sample provenance and regeneration assets are available.
Restricted or separately supplied packets	Stress tests unless public row binding and source-confounding checks pass.

source-documented point estimate. The consumer-boundary gate requires the artifacts needed by the named consumer to be visible before wording is assigned. The surface-delta gate checks whether a claimed second asset actually differs in model, data, modality, or response contract instead of repeating the same family. Each gate corresponds to a failure mode observed in the boundary cases: unbound checkpoint claims, source-pool shortcut membership, code-only or copied-metric promotion, high-score claims without an action-compatible consumer surface, and adjacent controls rewritten as portability evidence.

Consumer requirements are fixed before gate labels are assigned. The only consumer templates are model-owner risk reporting, replay audit, and downstream summary; Table II records the artifacts each task must expose. A consumer-boundary failure means the named task cannot be performed from the packet.

Code, checkpoints, score files, and high candidate scores may still fail a downstream claim when target, split, metric, consumer, or surface-delta surfaces are absent. The contract asks which sentence an artifact supports, not whether the artifact has other scientific value or fully reproduces the original paper.

Low-FPR metrics are reported as finite empirical packet readouts [13]. For example, TPR@0.1%FPR on 100 nonmembers has a false-positive budget of 0.1, so zero observed false positives are allowed under the strict empirical rule. Describing a small-packet tail value as a calibrated continuous sub-percent operating point would overstate the evidence. Many diffusion MIA packets are small, and strict-tail claims can otherwise sound stronger than the evidence permits.

Gate labels are claim-relative measurement decisions. *Pass* means the packet exposes the surface required for the stated claim: replay claims require row-bound score, response, or target-score arrays, while point claims require a frozen source-documented metric and narrower wording. *Partial* means a useful surface exists but cannot support a stronger claim without more identity, provenance, or consumer binding; *Fail* means a missing or contradictory surface blocks that claim; and *N/A* means the gate is outside the claim type. Each label records the missing surface that would change the state, so the protocol produces an audit decision, not a repository quality score. A metric gate pass is not a single replay claim: it must carry a replay tier, and point-estimate passes cannot be used for interval or dominance claims that require direct score arrays.

TABLE II
EVIDENCE-CONTRACT GATES AS PROTOCOL CHECKS.
CONSUMER-BOUNDARY ROWS MAKE THE DOWNSTREAM ACTION AND MISSING ARTIFACT EXPLICIT BEFORE WORDING IS ASSIGNED.

Gate	Required surface checked by the protocol
Target identity	Fixed architecture, checkpoint, training provenance, or target descriptor.
Split semantics	Immutable member/nonmember row manifests or split definition.
Score/response coverage	Row-level scores, responses, features, or metric packet for labeled samples.
Metric provenance	Replay tier: row-score, response/feature, target-score, or documented point estimate.
Consumer boundary: model owner	Can name target/checkpoint, split, access mode, operating point, finite-tail denominator, and owner action for the row role.
Consumer boundary: replay auditor	Can recompute or verify from row-score, target-score, response, or feature packet plus metric code/JSON; source points stay point-only.
Consumer boundary: downstream report	Can quote target/split, replay tier, provenance, denominator, and caveat in a bounded sentence; hidden fields block audit wording.
Surface delta	Evidence that a claimed second asset changes model, data, modality, or response contract and is not the same adjacent surface.

Consumer-boundary labels therefore block claims that require hidden data, new execution, or a different evidence surface.

V. MEASUREMENT PROTOCOL

We assign each result to one of five evidence states. A reportable row is eligible for contracted reporting only under its stated boundary and caveat. A candidate row can support follow-up analysis but is not report-ready. A support-only row can motivate or contextualize a mechanism but cannot serve as a direct benchmark. A watch item is tracked for future artifacts. A negative row answers a bounded decision question and usually closes a low-value expansion.

The protocol is a finite-state decision rule over a proposed audit claim and its supporting packet. Its input is an audit claim template c and an evidence packet $p = (T, S, O, M, C, \Delta)$ binding target T, split S, row-level observable O, metric provenance M, consumer boundary C, and optional surface delta Δ. It evaluates the ordered gate vector $g(p, c) = \langle g_T, g_S, g_O, g_M, g_C, g_\Delta \rangle$ and transitions to the narrowest state justified by the first required non-Pass gate. The pair (p, c) is reportable only when target, split, evidence, metric, and consumer gates pass for the proposed wording and the delta gate either passes or is not applicable. A reportable row is *replay-admitted* only when its metric tier exposes row, target-row, response, or feature replay; a frozen source-documented point estimate can be reportable, but only for exact bounded point wording. Figure 1 makes the transition rule explicit.

The coded claim register is the manuscript state table: each checked claim maps to one first blocker, state, replay tier, and allowed sentence. It blocks unsafe dominance, interval, portability, prevalence, and route-generalization wording when

Gate-assignment procedure.
1) Bind T, S, O, M, C, Δ from public packet surfaces and source provenance before reading aggregate counts.
2) Evaluate gates in fixed order: target, split, evidence, metric, consumer boundary, and delta. Assign Pass only when the surface needed by claim c is visible in the packet; assign Partial when a related surface is visible but cannot support the proposed wording; assign Fail for missing or contradictory required surfaces; assign N/A only when the claim type does not use that gate.
3) Record the first non-Pass required gate as the first blocker. Later passes do not override this blocker.
4) Emit the narrowest allowed state: reportable for all required Pass gates, candidate for unresolved consumer or portability gates, support-only for different access or feature surfaces, watch for artifact discovery without row binding, and negative for a bounded route-selection stop.
5) Attach replay tier. Row-score, target-score, response, or feature replay may be replay-admitted; source-documented point estimates remain point-worded even when reportable.

Fig. 1. Fixed-order gate-assignment procedure used by the coded claim register.

prose exceeds the recorded state or provenance. This validates release-scope claim consistency for selected boundary cases; it does not establish that the six gates are necessary or sufficient for all diffusion MIA audit correctness, nor does it estimate independent-coding claim-change rates. The supplement also exports `claim_transition_examples.csv`, a four-row reader-facing slice derived from the trace, so the reportable, candidate, uncertainty-sidecar, and C14 packet-ready cases can be checked without opening the full register.

Metrics are computed from row-level scores or frozen response packets; otherwise they are retained as declared point estimates. We report AUC, attack success rate (ASR), TPR@1%FPR, and TPR@0.1%FPR. Low-FPR values are interpreted against the nonmember denominator. The implementation takes the maximum empirical TPR at thresholds with FPR at most the target; when $0.001n_0 < 1$, TPR@0.1%FPR is a zero-FP tail, not a continuous operating point. The trace records both metric derivations and packet boundaries.

A. Metric Replay

Metric replay is a provenance check. When a score array exists, metrics are recomputed from row labels instead of copied from prose. When only responses exist, the scorer and response manifest must be fixed before the metric can be used. For a nonmember packet of size n_0, FPR takes values on a discrete grid of $1/n_0$. Thus TPR@0.1%FPR is a strict empirical tail whenever $0.001n_0 < 1$. We still report the readout because it is useful for comparing packets, but the text must retain the denominator and avoid calibrated-risk language [13].

Table III is used later as a claim-admission diagnostic: rows that look usable under ordinary code-or-metadata availability are blocked by different missing gates once row binding, metric replay, consumer boundary, and surface delta are evaluated.

The matrix shows which missing surface blocks score-only, code-only, and replay-only claims. The contract ties each allowed sentence to the first blocker, replay tier, claim trace, source provenance, and metric-uncertainty sidecars.

VI. Boundary-Case Trace and Claim Outcomes

For selected boundary cases, we apply the coded gates and record the evidence state plus diagnostic tag: reportable, candidate, support-only, watch, or negative, with known-surface miss, source-confounded, or metadata-only where applicable. We also mark rows where weak admission rules would admit a claim that the contract blocks. Table IV fixes the selected row sets before counts are read. The declared set is a boundary-case stress suite, not a rate estimator: it tests claim-state behavior across bounded positives, a high-score metric-only non-admission case, route closures, and L0/L1 artifact stops. The contract would require revision if a blocked row had fixed target identity, fixed membership semantics, row-bound scores or responses, replayable metrics, and a compatible consumer boundary, or if the first blocker could not be reproduced from the packet.

Rows enter when they instantiate at least one gate-relevant surface or blocker before aggregate counts are read. The design has five control roles. Reportable rows are replay/source-tier anchors. H2 is a metric-strength boundary case: high AUC with failed consumer and delta gates. ReDiffuse STL-10, CommonCanvas, and MIDST are route-closure checks. Tracing the Roots and the separately supplied Stable Diffusion packet test support-only and source-confounded states. Fixed-search, source-query, and targeted-link rows test row-set discipline and L0/L1 artifact stops.

Rows entered the declared set by rule before count readout. We included decision-affecting artifacts with an evidence source, released trace entry, or machine-readable artifact; recorded duplicates as deduplication or screening rows; and excluded background literature, artifact-discovery notes, large new-execution assets, and citation-only rows. Later label changes are reported as label-review resolutions or source-query control updates, not unreported exclusions. The selected rows show how the protocol accepts, narrows, or blocks fixed claim wordings; they are not a sampling frame for external row frequencies.

Each C1–C14 claim state has an inspectable first missing surface in the coded claim register. For every C1–C14 claim, the trace and provenance sidecars list the gate labels, first blocker, replay tier, state, allowed wording, source hashes, commit identity, generator command, and availability tier. Asset generation checks that provenance IDs resolve and gate labels use allowed values. For the current packet, these fields

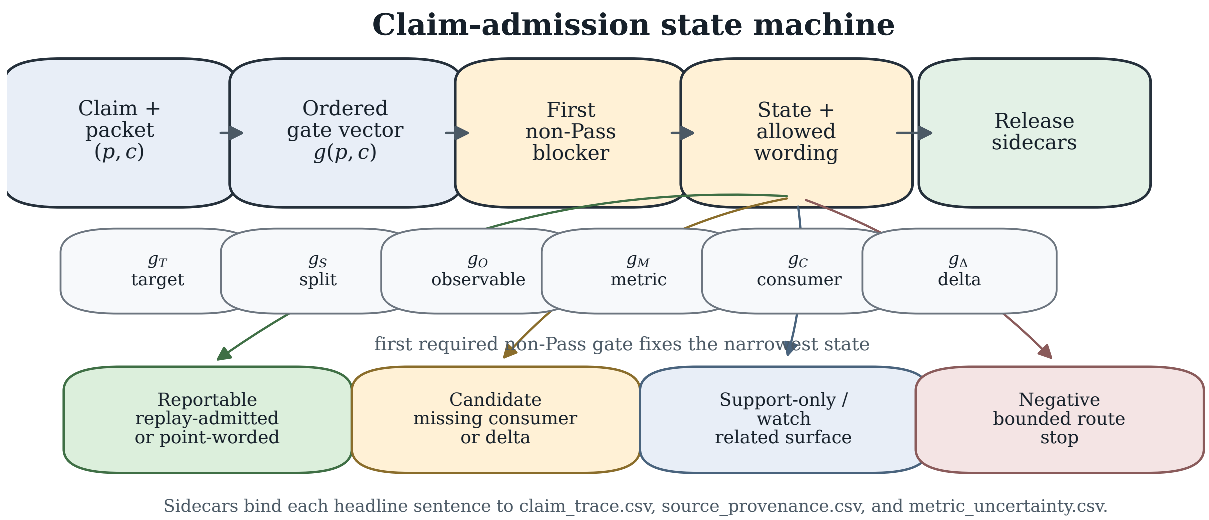

Fig. 2. Claim-admission state machine. The ordered gate vector and first required non-Pass blocker route a proposed claim to reportable, candidate, support-only/watch, or negative wording; release sidecars bind the resulting sentence to trace, provenance, and metric files.

TABLE III
CLAIM-ADMISSION GATE MATRIX EXAMPLES. LABELS ARE CLAIM-RELATIVE: A ROLE-PASS IS SUFFICIENT FOR THE STATED PAPER ROLE, NOT AUTOMATIC DOWNSTREAM ADMISSION. REPORTABLE ROWS ARE SPLIT BY REPLAY TIER SO SOURCE-DOCUMENTED POINT ESTIMATES ARE NOT CONFLATED WITH ROW-SCORE REPLAY.

Route	Target	Split	Evidence	Metric	Boundary	Delta	Paper use
PIA rows (row replay)	Pass	Pass	Pass	Pass	Pass	N/A	Row-replay gray-box audit and defense-comparator rows
Recon/GSA rows (point tier)	Pass	Pass	Point	Point	Pass	N/A	Black-box proxy and white-box upper-bound comparator rows
DPDM W-1 bridge (target-score replay)	Pass	Pass	Pass	Pass	Pass	N/A	Target-score defense bridge
H2 output-cloud	Pass	Pass	Pass	Pass	Fail	Fail	Candidate only; consumer and portability blocked
SD/CelebA img2img portability	Pass	Pass	Pass	Pass	Fail	Pass	Candidate boundary; not portability admission
MoFit public score files	Partial	Partial	Partial	Partial	Fail	Fail	Support-only public score-file boundary; row binding and consumer blocked
ReDiffuse STL-10 strict/overfit	Partial	Pass	Partial	Partial	Fail	Fail	Strict route negative; overfit support-only
CommonCanvas response packet	Partial	Pass	Pass	Pass	Fail	Pass	Bounded negative
Tracing the Roots feature packet	Partial	Pass	Pass	Pass	Partial	Pass	Support-only
Supplied SD ReDiffuse	Partial	Partial	Pass	Pass	Fail	Pass	Source-confounded support

identify only the local review snapshot; dirty tree state and local paths are not clean public-release provenance.

A. Boundary-Case Trace

The declared set was built from the audited evidence surface and frozen before count readout; standalone aggregate claims would require a larger distinct-surface frozen corpus and an independent multi-reviewer label audit for coding reliability. The supplement includes a shuffled blank C1–C14 claim-gate recode template and hash manifest for possible future independent recoding. The template omits author gate labels; no recode labels or reliability result are reported in this paper. The release also includes a selected report-correctness fault-injection matrix for the admitted-only risk-card renderer and candidate-promotion phrase guard. Ten hand-authored cases check expected behavior for this renderer: one current-card acceptance, four renderer rejections (candidate insertion, missing finite-tail denominator, private source path, and row-count drift), three phrase flags (two direct candidate promotions plus metadata-drift-to-compute-release wording), and two clean-boundary controls. These tests exercise this renderer and phrase guard only; they do not estimate report-drift reduction or external-use behavior. We also ran a supplementary metadata search on 2026-05-26 over seven GitHub queries and five arXiv API queries, with no downloads or GPU runs. It records three re-found surfaces, one new code-public no-packet surface, seven relevant metadata-only rows, one existing-watch metadata row, four empty fixed-query rows, and one noisy query row; no observation changed an admission decision. These rows expose claim boundaries,

TABLE IV
FROZEN BOUNDARY-CASE LAYERS USED FOR CLAIM-STATE CHECKS.

Layer	Fixed row set and validation role
Boundary-case records	21 rows: 11 previously reviewed evidence-surface rows plus 10 metadata-only expansions; positive, negative, support-only, and source-confounded controls; no new downloads or model runs.
Fixed metadata batch	2026-05-26 pass with 7 GitHub repository queries and 5 arXiv API metadata queries; 17 rows test metadata-only L0/L1 stops and refound/noisy surfaces; no row changed an admission decision.
Source-query controls	2026-05-27 metadata-screening HF/Zenodo/OpenReview API pass; 10 observations test query-result hygiene and non-poolable row-set discipline; no observation changed an admission decision.
Targeted artifact links	2026-05-27 chase over existing metadata-only seeds; 3 L1 public artifact surfaces, including 2 split surfaces; artifact-inspection stops only, with no score/metric replay or row-set pooling.
Claim scope	Counts support boundary inspection for this declared set only; they are not field-prevalence, artifact-quality, or reproduction-rate estimates.

but they are metadata-only and are never pooled with replayed score packets. For trace inspection only, rows are labeled by claim-support level: L0 metadata discovery, L1 artifact inspection, L2 scoreable/replayed packets, and L3 consumer-boundary support. The 21-row selected-set counts are 8/3/9/1, and the 17-row metadata-screening fixed-search counts are 13/4/0/0. These are not prevalence or replayability estimates. Fixed-search rows provide discovery/inspection provenance only, not replayability or auditability rates. A targeted artifact-link sidecar adds three L1 public artifact surfaces, but no row-score, ROC, metric JSON, or verifier packet. Automated cross-reference of gate labels against trace entries checks internal release consistency; it is not independent validation. In this packet, that check found no invalid states or route-changing contradictions. An same-team second pass changed no admission state and tightened two labels (withdrawn DLM delta, Noise Aggregation target). Publishing standalone artifact claims would still need the matching evidence: a larger distinct-surface corpus for aggregate behavior, or independent label verification for coding reliability.

B. Author-Keyed False-Promotion Stress Controls

C14 addresses a narrower question than the admitted row protocol: when weak public-surface cues are present, which recorded contract gate still prevents an admission claim? The packet contains thirteen selected no-download E2 public-surface checks. A check is included only when a weak-admission trigger is present and the public surface lacks a target-bound score/response packet or mismatches diffusion-generator membership evidence. Table V summarizes the weak rules and author-keyed gate blockers; row-level examples remain in the packet. Fig. 3 visualizes the selected-row weak rules. No reviewer has labeled this packet, and no external-review or reliability result is reported; the status artifact records `n_reviewers=0` and `allowed_claim_scope=packet_ready_only`. The packet reports only weak-admission triggers and the first contract gate that blocks each row (first blocker). It is an author-keyed preparation packet, not admitted evidence.

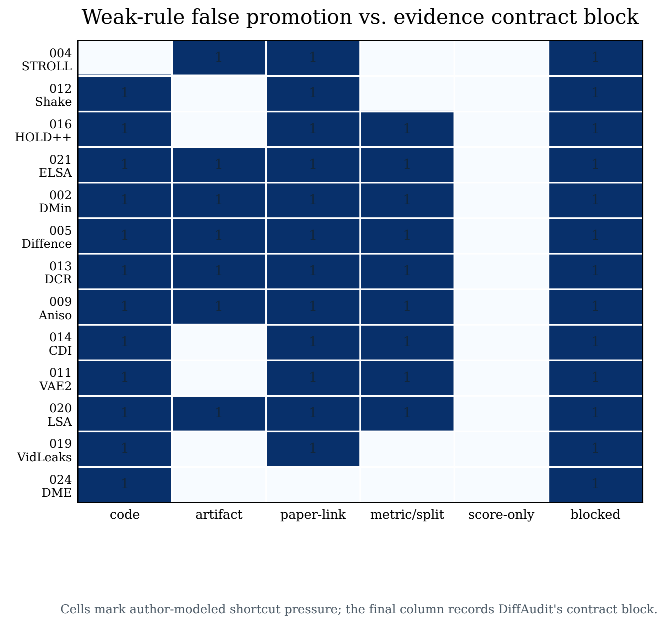

Fig. 3. C14 author-keyed stress visualization for selected rows. Cells mark weak-admission triggers before reviewer labeling; the final column records the author-keyed first blocker. This figure is a packet-readiness diagnostic, not reviewer evidence.

Across the 13 selected rows, weak-admission triggers appear as follows: paper-artifact-link in 12 rows, code in 12, metric/split in 9, artifact in 7, and score-only in 0. The contract-side block is also author-keyed: no selected row passes the score/response or consumer-boundary gate (13/13 recorded non-Pass), and target/metric have no author-keyed Pass labels.

Several C14 rows carry an author-keyed blocker because the artifact targets a different question: attribution, classifier defense, copying, prompt memorization, or dataset-level identification. External agreement and prevalence are out of scope for C14 and require a separate reviewer-label protocol and corpus design.

VII. CALIBRATION EVIDENCE BUNDLE

Rows enter the paper through a fixed calibration path. A calibration row must bind labeled samples to a stated target, expose metric provenance, attach finite-tail denominators, and carry consumer-boundary language. The bundle has two evidence strengths: PIA, PIA-dropout, and DPDM W-1 are replay-admitted from row or target-score arrays in the audited artifacts; recon and the main GSA row are source-documented point estimates, so they can support only their exact bounded role claims and not interval, dominance, or leaderboard claims. This is why bounded reportable roles can coexist with non-admitted candidates that have large scores: admission is about reusable audit semantics, not metric magnitude. The five cases are target/split specific: recon DDIM uses CIFAR-10 public-100 point evidence; PIA and PIA+stochastic dropout use

TABLE V

C14 WEAK-RULE COUNTS AND CONTRACT BLOCKERS. COUNTS ARE OVER THE THIRTEEN SELECTED E2 STRESS ROWS ONLY; `N_REVIEWERS=0` AND `PACKET_READY_ONLY`. THE TABLE REPORTS PRE-LABEL STRESS-CONTROL SIGNALS, NOT EXTERNAL ADJUDICATION, PREVALENCE, OR ADMITTED EVIDENCE.

Weak rule	Rows triggered	Contract-side blocker	Allowed claim
Code availability	12/13	Rows are recorded non-Pass at the score/response and consumer-boundary gates (13/13); runnable code alone is not credited with binding target rows, responses, scores, or verifier.	Pre-label stress-control only.
Paper-artifact link	12/13	Public links are not credited as an admitted row-bound score/response packet or verifier.	Pre-label stress-control only.
Metric/split visibility	9/13	Target and metric gates receive no author-keyed Pass labels; split hints or metric code are incomplete without row-bound outputs.	No downstream audit claim.
Artifact availability	7/13	Artifact surfaces are not credited in this packet with binding target identity, split, response/score provenance, and consumer boundary at once.	Pre-label stress-control only.
Score-only	0/13	No public target-bound score/response packet is available in the selected stress rows.	No score-only admission.

CIFAR-10 512/512 row replay; GSA uses CIFAR-10 1k/1k point evidence; and DPDM W-1 is target-score bridge replay.

Table VI summarizes five worked-example cases for the admission protocol. The five cases cover different access modes: black-box reconstruction and gray-box PIA baseline provide primary leakage-risk rows under their stated modes, stochastic dropout and DPDM W-1 are reportable under defense-comparator or bridge wording, and GSA remains a source-documented upper-bound point comparator. Their shared unit is the evidence contract, not a common attack family or leaderboard rank. The table's provenance shorthand is part of the claim: *row* denotes row-score replay, *point* denotes source-documented point estimates without replay-derived intervals, and *target row* denotes target-score replay for the defended bridge. AUC uncertainty is reported only where the replay surface supports it: the companion metric table gives bootstrap intervals for PIA baseline [0.8168, 0.8640], PIA-dropout [0.8039, 0.8535], and DPDM W-1 [0.4640, 0.5152]. These intervals are AUC intervals under exchangeable-row bootstraps; they are not interval claims for TPR@1%FPR or TPR@0.1%FPR. Tail readouts remain denominator-and-count packet summaries. Recon and the main GSA row remain point estimates, and H2 intervals remain candidate-side controls. The compact paper table lists caveats and a finite-tail basis; the machine-readable bundle retains quality/cost metadata for replay. We do not pool these rows into a single leakage estimate; each row supports only the consumer claim named by its boundary or caveat.

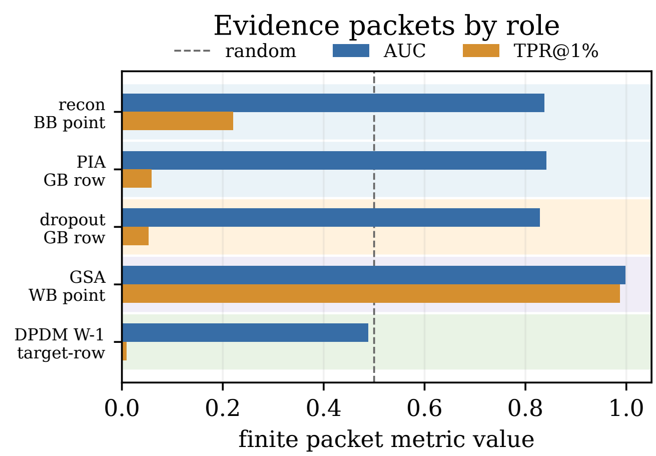

Fig. 4. AUC and TPR@1%FPR for role-separated worked-example cases. Rows are ordered by access and allowed claim role, and each label includes its replay tier. The figure is an inventory of role-separated evidence, not a metric-ranked benchmark leaderboard; plotted point rows do not carry replay-derived intervals.

VIII. METRIC-STRENGTH BOUNDARY CASE: HIGH AUC IS INSUFFICIENT

H2 is a metric-strength boundary case, not a new black-box attack. It satisfies same-family diagnostic checks but fails the consumer-boundary and surface-delta gates, so the packet remains candidate-only before the AUC is read. The failure criterion is fixed before the AUC readout: AUC 0.9615 still fails admission when consumer-boundary and surface-delta evidence are missing. The fixed scorer featurizes within-query response clouds from repeated H2 generations, fits a held-out logistic readout over output-output geometry, and excludes direct query-to-output distance. Its artifacts retain aggregate held-out prediction summaries, coefficients, and recorded intervals; no new responses are generated for this paper. Relative to black-box variation and reconstruction attacks that compare a query with returned or averaged outputs [19], [6], H2 removes query-to-output distance and scores output-output geometry. We treat it as a response-contract observable probe whose admission state remains candidate-only.

Metric strength alone is insufficient for admission. We include H2 as a high-AUC non-admission case: on the existing 512/512 cache, output-cloud logistic scoring reaches AUC 0.9615, ASR 0.9004, TPR@1%FPR 0.3340, and

TABLE VI
CALIBRATION EVIDENCE PACKETS WITH BOUNDED CLAIM ROLES. EACH ROW SPECIFIES THE EVIDENCE TIER AND METRIC PROVENANCE. BOTH LOW-FPR COLUMNS ARE FINITE-TAIL PACKET READOUTS TIED TO THE NONMEMBER COUNT n_0; TPR@0.1%FPR ALSO REPORTS RECOVERED TRUE POSITIVES OVER MEMBER COUNT AND THE FALSE-POSITIVE CAP IMPLIED BY n_0. POINT ROWS REPORT SOURCE-DOCUMENTED VALUES; REPLAY-DERIVED INTERVALS ARE REPORTED IN `METRIC_UNCERTAINTY.CSV` AND ADJACENT TEXT, OUTSIDE THE COMPACT TABLE.

Role/tier	Method	AUC	ASR	TPR@1%FPR (n_0 tail)	TPR@0.1%FPR (TP/mem; FP cap)	Claim role
BB main / point	recon DDIM public-100 step30 [6]	0.837	0.740	0.220	0.110 (11/100; cap 0)	black-box proxy point estimate; no row-score replay or downstream claim
GB main / row	PIA GPU512 baseline	0.841	0.786	0.059	0.012 (6/512; cap 0)	bounded gray-box row replay with provenance caveat
GB comp. / row	PIA + stochastic dropout	0.828	0.768	0.053	0.010 (5/512; cap 0)	provisional defense comparator
WB comp. / point	GSA 1k-3shadow [7]	0.998	0.990	0.987	0.432 (432/1000; cap 1)	upper-bound white-box comparator
WB bridge / target	GSA against DPDM W-1 [7], [22]	0.489	0.499	0.009	0.000 (0/1000; cap 1)	target-score defense bridge

TABLE VII
H2 OUTPUT-CLOUD ADMISSION-DECISION SUMMARY.

Question	Evidence	Contract decision
Metric signal	Output-cloud AUC 0.9615; raw/lowpass H2 diagnostics AUC 0.9057/0.8957.	Candidate observable only; metric strength is insufficient.
Same-family controls	Label-shuffle AUC 0.5076; shared-position, seed, and same-family cache-reuse AUC 0.9489–0.9705.	Same-family reuse support only; not portability evidence.
Admission boundary	SD/CelebA img2img 25/25 AUC 0.7888 with zero strict-tail recovery ($n_0 = 25$).	Insufficient for portability admission from the current packet; no downstream audit admission path is exposed.

TPR@0.1%FPR 0.1172 (60 true-positive members out of 512 at zero false positives). Table VII reflects the admission outcome: the portability packet and consumer-boundary admission path remain insufficient with AUC 0.9615 and same-family controls.

The portability boundary is the admission decision. On the SD/CelebA img2img 25/25 portability-test cache, output-cloud geometry reaches only AUC 0.7888 with zero strict-tail recovery ($n_0 = 25$). A smaller 10/10 stability cache is diagnostic only and remains too small to support portability evidence. Output-cloud geometry is reported only as a controlled response-cloud observable in a candidate-only boundary case: label shuffles collapse to random-level AUC as a sanity check, shared-position and same-family cache checks support only same-family stability, and the img2img packet is insufficient for portability or downstream audit admission.

IX. NEGATIVE AND SUPPORT EVIDENCE

Bounded non-admitted routes are retained only when they answer a route-selection question. Table VIII records the packet or provenance stop, first blocker, and conclusion that the record forbids; score-bearing rows and metadata-only route stops are not pooled as one metric corpus. The routes cover ReDiffuse [19], the CommonCanvas response surface associated with CopyMark materials [23], MIDST [24], and Tracing the Roots [18], plus MoFit, inspected only through public score-like text files as a support-only boundary check [20]. For STL-10, a generated route sidecar separates the strict 50k/50k DDIM reproduction rows from the fixed-subset overfit setting: the strict rows remain bounded-negative, while the overfit rows are support-only memorization sanity evidence.

A. Public Score-File Boundary: MoFit

MoFit gives a public score-file boundary because the checked COCO text files are score-like and the paper-side recomputation has a high AUC, but the row remains non-admitted. Over 500 train and 500 test positions, the best check ($\alpha = 0.55$) reports AUC 0.941948, ASR 0.883, TPR@1%FPR 0.488, and TPR@0.1%FPR 0.324; permutation labels are random-level (mean AUC 0.500805, 95% interval [0.465629, 0.535924]). The public target is local-path identified, score rows lack explicit row IDs, caption-order binding is only DiffAudit-side support, and no official row-bound metric JSON, ROC, verifier, or surface-delta control is observed. All six gates remain non-Pass, so MoFit is not C14, not N50 denominator evidence, not admitted evidence, not a second public asset, and not a compute-release target.

The decision value is route-specific. Bounded negative routes close tested routes; they do not refute the original papers. Positive feature packets remain support-only, source-confounded packets become boundary stress tests, and metadata discovery records decision provenance without broad prevalence inference.

X. DISCUSSION AND LIMITATIONS

A diffusion MIA score is interpretable only when the packet fixes target identity, split semantics, row coverage, metric provenance, finite-tail denominator, and consumer boundary. Without those surfaces, high AUC can reflect source-label shortcuts, prompt-conditioned auxiliary features, non-replayable scores, or finite-tail artifacts. H2 therefore remains candidate-only without a non-adjacent response asset and

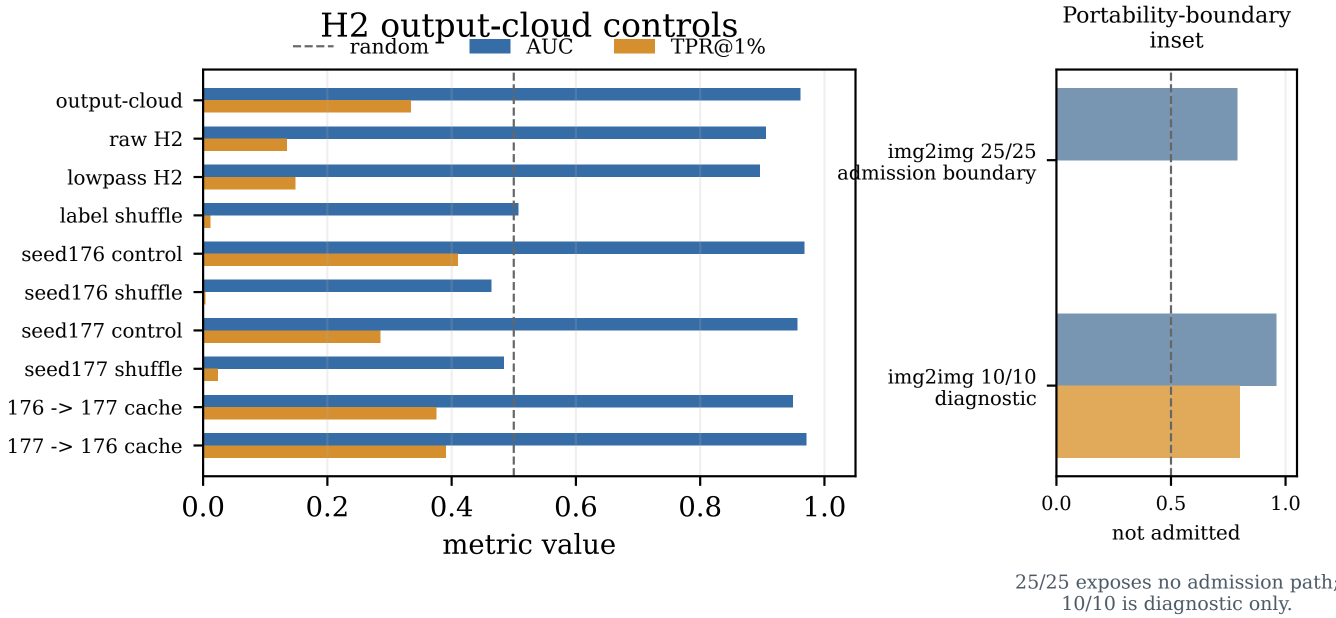

Fig. 5. Candidate-side failed admission stress test for H2 output-cloud geometry. The H2-family metric is above chance but remains non-admitted; it collapses under label shuffle and persists under seed-offset control, seed stability, and same-family cache-reuse checks. SD/CelebA img2img portability rows are shown as a boundary inset: the 25/25 packet does not expose an admission path, and the 10/10 row is diagnostic only.

TABLE VIII
NEGATIVE/SUPPORT PACKETS ARE ROUTE-LOCAL CLAIM DECISIONS.

Route	Packet/readout	First blocker	Not allowed
ReDiffuse STL-10	Strict 300K/500K rows stay random-level (AUC 0.5025–0.5140); fixed-subset overfit reaches AUC 0.9987–1.0.	Strict route is negative; overfit changes target, split, and training surface.	Strict reproduction success or general method refutation.
CommonCanvas	50/50 response packet; denoising-loss AUC 0.5148.	Route-local negative; no portable second-contract evidence.	Portable second asset.
MIDST	Best available score AUC 0.5981.	Tabular benchmark surface and weak image-diffusion relevance.	Image-diffusion audit evidence.
Tracing the Roots	Feature packet AUC 0.8158.	Feature tensor only.	Raw image audit row.
MoFit COCO	Support-only score-file check: all six gates remain non-Pass; AUC 0.9419 and TPR@1%FPR 0.488.	Numeric rows lack explicit row IDs and immutable public target binding.	Second public asset or admitted row.
Separately supplied SD ReDiffuse	Replay AUC 0.7103; source-only AUC 1.0.	Source-label separation.	Same-distribution proof.
Fixed-search trace	17 metadata observations; no evidence+metric pass.	Metadata-only protocol.	Literature-wide rate.
Source-query controls	10 source-query observations; not pooled with corpus counts.	Row-set control only.	Field-coverage rate.

consumer-boundary admission path. MoFit improves public score-surface stress coverage, but it does not repair immutable target identity, explicit row IDs, certified row binding, consumer-boundary, or surface-delta gates; it is not a second admitted asset. Selected-corpus, coding-reliability, and point-role claims likewise require a larger frozen corpus, independent multi-reviewer labels, or row/target-score arrays.

A. Release and Validity Threats

Reusable diffusion MIA artifacts should expose target/checkpoint identities, immutable splits, row-level scores or responses, metric code or JSON, finite-tail denominators, hashes or commit IDs, and intended consumer language. The reportable bundle is limited by permissions and available assets: rows come from local score/response replay or source-documented point estimates, not full upstream training-pipeline reproduction. We claim audit replay at the exposed surface, not universal leakage or method dominance. Low-FPR tails remain finite empirical packet readouts tied to nonmember denominators, and the fault-injection cases exercise only the release renderer and phrase guard.

The protocol checks surfaced evidence, not representativeness or completeness. A packet could pass all six gates while still misleading if the split is easy, the observable is chosen after inspection, or a known confound is omitted. When a packet is shared, include the claim trace as an immutable snapshot: a reviewer maps each headline claim to its trace row and checks the source-provenance hashes, generator command, commit identity, and availability tier. In the current packet, the provenance table is a local packet-identity snapshot; dirty tree state and local paths identify the review snapshot and must not be read as clean public release provenance.

Gate labels are same-team protocol labels over selected decisions, not inter-rater measurements or prevalence samples. They cannot substitute for independent recoding. C14 is a thirteen-row internal stress packet; field claims require a preregistered larger corpus and independent labels.

XI. Conclusion

This paper treats diffusion MIA evidence as bound to the claim it can support. In these boundary cases, the contract admits replay and point rows only under bounded wording, keeps H2 candidate-only, treats MoFit as a support-only public score-file boundary check, and leaves C14 as a packet-ready stress control rather than admitted evidence. The reporting rule follows from those gates: name the target, split, row-level observable, metric provenance, finite-tail basis, and consumer boundary, or report only the narrower claim supported by the packet.

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